

**Gradient Conventionalization: A Unified Framework for Lexical Semantics and Social
Epistemology**

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Abstract

This paper proposes a unified framework that reconceives the relationship between lexical meaning and world knowledge as a gradient of conventionalization rather than a categorical distinction. Meaning components are distinguished by their level of conventionalization, operationalized as degree of speaker agreement, rather than by membership in "lexical" versus "encyclopedic" categories. The framework introduces a five-level gradient — reference, denotation, sense, cultural connotation, private connotation — corresponding to decreasing intersubjective agreement, and a typology of entailments (strong, default, weak) correlated with these levels. This reconceptualization clarifies semantic prosody, explains why some arguments converge while others polarize, and connects lexical semantics to social epistemology. Central to the epistemological application is the diagnosis of *alpha misattribution*: the error of treating a term whose meaning is genuinely low-agreement as though it carried the stability of a high-agreement term, an error that converts unresolved conceptual disagreement into the appearance of bad-faith resistance to consensus. The framework generates testable predictions about semantic change, argument dynamics, and the conditions under which making agreement levels explicit can reduce polarization.

Keywords: lexical semantics, conventionalization, semantic prosody, entailment, intersubjective agreement, epistemology of disagreement

Gradient Conventionalization: A Unified Framework for Lexical Semantics and Social Epistemology

What words mean and what speakers know about the world have long been held apart by a boundary that formal semantics needs and cannot secure. On one side sits lexical knowledge: systematic, compositional, shared. On the other sits encyclopedic knowledge: contingent, individual, pragmatic. The distinction explains why some inferences behave like entailments and others like defeasible expectations, and why theories of meaning can be bounded at all rather than swallowing the whole of what speakers happen to know. But the boundary itself has never held still. Corpus linguistics has documented statistical regularities in word use — the tendency of *cause* to collocate with negative outcomes, of *regime* with authoritarian governments — that are too systematic to dismiss as noise and too easily overridden to count as classical entailments (Louw, 1993; Sinclair, 1991; Stubbs, 2001). Theorists disagree, framework by framework, about which side of the lexical-encyclopedic line these patterns fall on, and the disagreement has proven durable rather than resolvable.

This paper argues that the durability is itself diagnostic. The lexical-encyclopedic distinction is not a boundary between kinds of knowledge but a gradient of conventionalization, and both what we call lexical meaning and what we call world knowledge are, at bottom, learned and conventional, differing only in how widely and stably they are shared across a speech community. What counts as "lexical" is simply the region of the gradient where agreement approaches unanimity; what counts as "encyclopedic" is the region where it does not. This single move dissolves a problem that has been chronically mishandled in the treatment of connotative meaning specifically. Prior work conflates two phenomena under one label — *cultural connotation*, the systematic, community-level patterning that generates real corpus regularities, and *private connotation*, the idiosyncratic personal association that generates none — and having

conflated them, concludes that connotation as a whole falls outside the reach of semantic theory. That conclusion follows only for private connotation. Cultural connotation is conventionalized enough to support theoretical treatment; it has simply been treated as though it belonged to the same category as memories of one's childhood dog.

The reconceptualization required is not confined to meaning; it extends to what meaning licenses. Classical entailment is categorical — either p entails q or it does not — and this categorical apparatus is the right tool for high-agreement meaning components but the wrong one for the moderate-agreement components that generate systematic collocational skew without producing anything like logical necessity. A typology of entailments graded by conventionalization level — strong, default, weak — is needed to do justice to the middle of the gradient, and once that typology is in place, semantic prosody stops looking like a puzzle: it is what cultural connotation looks like when it generates weak entailments that interact with discourse pragmatics.

The same gradient, I will argue, does explanatory work well outside lexical semantics. Whether an argument converges toward consensus, stalls in unproductive impasse, or actively polarizes depends on the conventionalization level of the terms the argument is conducted in. This is not a metaphorical extension of the semantic apparatus; it follows from the same mechanism that generates weak entailments in the first place. Terms at high conventionalization support shared inference rules, which is why disputes about whether a given creature is a dog get resolved by looking at the creature rather than by negotiating what *dog* means. Terms at moderate conventionalization do something more specific and more consequential than merely fail to converge: they invite a particular cognitive error, in which participants treat a term that is genuinely low-agreement as though it possessed denotation-level stability, and read the other side's failure to agree as bad faith rather than as the predictable consequence of arguing in low-

agreement vocabulary. I call this error *alpha misattribution*, and I will argue that it, rather than low agreement on its own, is what converts unresolved conceptual disagreement into polarization. The paradigm case developed below is the debate over abortion, not because that debate is uniquely intractable but because it is uniquely clear about where the misattribution occurs and what would be required to correct it — a correction that is epistemic, not moral, since recognizing that a term is currently low-agreement is compatible with any view about whether the matter has an underlying fact of the matter at all.

The paper proceeds in eight sections. The next section examines why the lexical-encyclopedic boundary has proven unstable across theoretical frameworks and empirical domains. Section three develops the gradient itself. Section four introduces the entailment typology. Section five diagnoses the conflation of cultural and private connotation. Section six applies the framework to semantic prosody. Section seven develops the epistemological application, including alpha misattribution and the abortion case. Section eight closes with predictions, limitations, and a brief synthesis.

The Instability of the Lexical-Encyclopedic Boundary

The distinction between linguistic and encyclopedic knowledge has organized formal semantics since Katz and Fodor's (1963) foundational work: linguistic knowledge is systematic and shared, encyclopedic knowledge is what speakers happen to know about the world, and semantic theory is bounded by restricting itself to the former. The distinction is intuitive at the extremes — that *bachelor* means "unmarried adult male" seems linguistic; that bachelors tend to eat takeout seems encyclopedic — and it becomes unstable almost immediately once pressed. Fillmore's (1982) analysis of *bachelor* is the classic demonstration: the word resists definition by necessary and sufficient conditions, since a Catholic priest and the Pope are unmarried adult males who are nonetheless not called bachelors, and correct application depends on cultural

knowledge about marriage practices and life-stage expectations that no theory has managed to relegate cleanly to either side of the boundary. The instability is not confined to one lexical item. Generative semanticists argued for incorporating more world knowledge into semantic representation (Lakoff, 1971) while formal semanticists sought a stricter boundary (Katz, 1977); cognitive linguists rejected the distinction outright, holding that all meaning is encyclopedic (Croft & Cruse, 2004; Langacker, 1987); frame semantics held that even paradigmatically lexical items require structured background knowledge to apply at all (Fillmore, 1982). If the boundary reflected a real division in the structure of meaning, theoretical frameworks converging on where to draw it would be expected; instead, they locate it at radically different points or abandon it.

Psycholinguistic evidence does not recover a categorical distinction the theoretical disputes fail to settle. Semantic priming shows both definitional features and encyclopedic associations facilitating processing along a continuous rather than a stepped function, with priming strength tracking frequency and typicality rather than membership in one knowledge type or the other (McRae et al., 2005). The N400 component, which indexes difficulty of semantic integration, shows graded sensitivity to violations of definitional features and violations of typical associations alike, with amplitude differences reflecting degree of expectation violation rather than a categorical type difference between them (Federmeier & Kutas, 1999; Kutas & Federmeier, 2011). Attempts to locate distinct neural substrates for linguistic versus general conceptual knowledge have largely failed to find them, with extensive overlap in the systems supporting both (Binder et al., 2009). None of this is what a real categorical boundary should produce.

Cross-linguistic variation compounds the problem rather than clarifying it. Color-term boundaries vary systematically but not arbitrarily across languages (Berlin & Kay, 1969) — too regular for pure encyclopedic contingency, too variable for a fixed conceptual structure that

language merely labels. Evidentiality is grammatically obligatory in many languages and optional, pragmatically inferred in English (Aikhenvald, 2004), yet the underlying distinction between direct and indirect evidence is the same content in both cases; whether it counts as lexical or encyclopedic tracks the accident of grammaticalization, not any deeper fact about the content itself. If conventionalization level, rather than kind of knowledge, is what varies across languages, this pattern is expected rather than puzzling.

Acquisition evidence tells the same story from a different angle. Children learn word meaning through the same exposure and the same statistical extraction that they use to learn facts about the world, with no evidence of a separate mechanism reserved for properly linguistic content (Tomasello, 2003). The acquisition of *bachelor* itself illustrates this: children first grasp the core reference, then gradually refine understanding of social eligibility and contextual restriction, and nothing in the developmental trajectory marks a transition from acquiring a lexical core to acquiring an encyclopedic periphery (Carey, 2009; Clark, 2009). A single continuous learning process producing a graded outcome is exactly what the gradient view predicts and the categorical view has to explain away.

The success of large language models supplies a final and, for present purposes, especially direct line of evidence. These systems are trained on next-word prediction over raw text, with no architectural distinction built in between linguistic and world knowledge, and they nonetheless acquire both syntactic competence and extensive factual knowledge through the same mechanism operating on the same distributional evidence (Brown et al., 2020). If lexical and encyclopedic knowledge required different learning mechanisms, this outcome would be mysterious; instead it suggests that distributional patterns in language encode both in a unified way (Bender & Koller, 2020). This should not be overstated into a claim that distributional exposure alone settles what understanding consists in — that is a separate and much larger

question, addressed at length elsewhere (Søgaard, 2022; Potts, 2019) — but the narrower point holds regardless of how that larger question is resolved: whatever mechanism a model uses to generalize, the fact that a single training signal supports both linguistic and world knowledge is evidence that the signal does not itself respect a lexical-encyclopedic partition, because no such partition was built into how the signal was used.

Taken together, these five lines of evidence — theoretical disagreement, psycholinguistic continuity, cross-linguistic variation, unified acquisition, and computational learning — converge on the same diagnosis. The boundary is not a natural division that theory has simply failed to locate correctly; it is an artifact of insisting that a continuous property be treated as a categorical one. What is needed is not a better boundary but a framework that replaces the boundary with the gradient it was always approximating.

Meaning as Graded Conventionalization

If the lexical-encyclopedic boundary is not a natural division, an alternative is needed that preserves what the boundary was tracking — that some aspects of meaning are more stable and more widely shared than others — without forcing that difference of degree into a difference of kind. The alternative proposed here treats meaning as existing on a continuous gradient of conventionalization, where conventionalization is operationalized as the degree of intersubjective agreement among speakers of a language community.

The foundation is the arbitrariness of the linguistic sign. Saussure's (1916/1959) observation that the pairing of a sound sequence with a meaning is a matter of convention rather than natural necessity is usually taken to apply narrowly, to the relation between a word-form and its referent. But the same arbitrariness runs through the internal structure of what a word means, not merely through the pairing of word and meaning. That English *bachelor* encodes "unmarried" and "adult male" as central while leaving desirability or living arrangements

peripheral is itself conventional; a different language could carve the same conceptual territory differently, and many do. If the internal structure of meaning is conventional throughout, then there is no principled place within it to insert a categorical boundary between a lexical core and an encyclopedic periphery. What we call lexical knowledge is simply the region of this conventional structure where agreement across the community approaches unanimity; what we call encyclopedic knowledge is the region where agreement is lower and more variable. Both are learned; both are arbitrary in Saussure's sense; they differ only in degree of conventionalization, not in kind.

Let a speech community be denoted $\mathcal{S}$, and let a semantic feature that some speakers associate with a word w be denoted f . An *agreement score* $\alpha(f)$ is the proportion of speakers in $\mathcal{S}$ who associate f with w , so that α ranges over the interval from 0, where no speaker makes the association, to 1, where every speaker does. This is the only formal apparatus the framework strictly requires, and it is worth being explicit about what it does and does not commit the theory to. It does not commit the theory to a single method of measurement: α can in principle be estimated through explicit judgment tasks, through the consistency of distributional patterns across corpora, through priming effects, or through the behavior of models trained on naturalistic data, and these measures should be expected to converge only approximately rather than identically, since they probe the same underlying disposition through different instruments. Nor does it commit the theory to treating any particular α value reported in what follows as an established finding rather than a theoretically motivated estimate awaiting confirmation by the relevant experimental or corpus methods; where specific values are mentioned below, they should be read in that light throughout.

For expository purposes, five regions of the gradient are distinguished, corresponding to points that recur in the semantic and lexicographic literature under separate names. *Reference* (α

close to 1) is the basic extensional mapping from a word to what it applies to — that *dog* applies to canines and not to chairs is agreed by virtually every competent speaker, and the rare disagreements concern genuine borderline cases such as wolves and foxes rather than variation in conventionalization itself. *Denotation* (α above roughly 0.90) comprises the core definitional features dictionaries treat as necessary conditions — that a bachelor is unmarried, that killing causes death — where agreement is very high but occasionally negotiable in extended or metaphorical use. *Sense* (α in roughly the 0.70 to 0.90 range) comprises the relational features that position a word within its semantic field: how *bachelor* contrasts with *married man*, how *knowledge* relates to *belief* and *justification*. Agreement here is still high enough to support communication but low enough that speakers organize the same semantic field differently, prioritizing different contrasts or different intuitions about borderline cases. *Cultural connotation* (α in roughly the 0.50 to 0.70 range) comprises the systematic evaluative and stylistic associations that corpus linguists have documented as semantic prosody — the association of *regime* with authoritarian governance, of *cause* with negative outcomes — shared by a majority of speakers and productive of real corpus patterns, but variable enough across speakers that a substantial minority lack the association or hold it only weakly. *Private connotation* (α below roughly 0.50, often approaching zero for highly specific cases) comprises the idiosyncratic associations of individual experience — that *hospital* evokes one speaker's memory of a parent's illness and another's memory of a child's birth — which are psychologically real for the individuals who hold them but generate no systematic pattern at the level of the community and cannot be recovered from distributional evidence, because they were never encoded in usage to begin with.

These five regions are reference points on a continuous scale, not discrete categories with natural boundaries between them, and the boundaries given above should be read as calibration

targets for the norming and corpus studies that would establish them precisely, not as measurements already in hand. Three properties of the gradient are worth stating explicitly because each does argumentative work later in the paper. Different meaning components of a single word can occupy different positions on the gradient simultaneously — the basic denotation of *bachelor* sits near the top while its associations with particular lifestyles sit much lower, and a word's meaning is properly represented as an ordered structure of such components rather than as a single point on the scale. Conventionalization level can shift over time through ordinary processes of semantic change, so that a feature beginning as private or cultural connotation can move toward sense or denotation as usage stabilizes, and a feature once denotational can weaken through bleaching. And the population-level agreement score for a term can diverge sharply from the agreement score computed within a subgroup of speakers who share a background theory, a professional training, or a political commitment — a divergence that will do substantial work in section seven, where it is what makes alpha misattribution possible in the first place.

What this framework buys, before any of its further applications are developed, is a principled replacement for a distinction that theory has needed and never been able to draw cleanly. There is no threshold in α that is theoretically privileged as the true boundary between lexical and encyclopedic; different theoretical traditions can be understood, on this view, as implicitly drawing that threshold at different points along a single continuous scale, which is precisely why they have never converged on where it belongs.

A Theory of Weak Entailments

If meaning exists on a gradient, the inferences meaning licenses should exist on a corresponding gradient of strength, and classical semantic theory, built around a single categorical notion of entailment, is not equipped to represent this. Classical entailment holds that

p entails q just in case q is true in every possible situation in which p is true; under this definition entailment is all-or-nothing, with no intermediate case. This is the right notion for the top of the conventionalization gradient — "John is a bachelor" entails "John is unmarried" with no room for exception — but it is the wrong notion for the systematic, corpus-documented patterns that motivated semantic prosody research in the first place. *Cause* appears overwhelmingly with negative complements — *problem*, *damage*, *concern* — yet "the policy caused economic growth" is neither contradictory nor especially marked (Stubbs, 2001). Treating this as pragmatic implicature, in the manner of Grice (1975), fails to capture how stable the pattern is across genres and corpora; ordinary implicatures are supposed to be far more context-sensitive and far more easily cancelled without residue than these patterns actually are. What is needed is a typology that recognizes gradations of entailment strength tracking gradations of conventionalization.

Three types are distinguished here. *Strong entailments* arise from features at the reference and denotation levels, where agreement is near-universal. These behave exactly as classical entailments do: *bachelor* entails *unmarried*, *kill* entails *cause to die*, and denying the entailment while asserting the premise produces contradiction or forces reinterpretation of the word in a nonstandard sense, as when *bachelor* is reinterpreted to mean an academic degree rather than an unmarried man. *Default entailments* arise from features at the sense level. These are inferences that hold in the absence of contrary information but can be explicitly overridden without contradiction: *bird* carries a default inference to flight, cancelled without anomaly in "the emu is a bird but cannot fly," yet the default remains active enough to prime flight-related processing even for speakers who know perfectly well that emus and penguins exist (McRae et al., 1997). The overridability without contradiction is what marks this as weaker than strong entailment; the persistence of the priming effect under cancellation is what marks it as stronger than mere

pragmatic suggestion. *Weak entailments* arise from features at the cultural connotation level. These generate systematic expectations that skew corpus distributions and influence real-time processing, yet they are cancelled without even the mild markedness that accompanies default-entailment cancellation: "the policy caused prosperity" and "the discovery caused excitement" pass without remark. What distinguishes a weak entailment from a private association that happens to correlate with usage is exactly the systematicity that private connotation lacks — a weak entailment shows the same moderate but consistent cross-speaker patterning that marks cultural connotation generally, where a private association shows none.

Cancellability, on this account, should track conventionalization level directly: the higher a feature's α , the harder it should be to deny that feature while retaining the word's ordinary sense, and the lower a feature's α , the more freely it should be deniable. This is stated here as a predicted monotonic relationship, to be established through acceptability-judgment studies of the kind proposed in section eight, rather than as a fixed mathematical function of α ; the earlier version of this framework proposed a specific linear formula for cancellability, and that specificity was unearned; nothing in the theory yet supports a claim about the functional shape of the relationship beyond its direction, and asserting a particular shape adds a commitment the framework cannot currently discharge.

Composition behaves differently at each level, and the differences follow directly from the entailment typology rather than requiring separate stipulation. Strong entailments compose the way classical entailments always have: if w_1 strongly entails f_1 and w_2 strongly entails f_2 , their combination entails both, which is why *female bachelor* is anomalous rather than merely surprising — the strong entailments of *female* and of *bachelor* are jointly unsatisfiable. Default entailments project through composition but yield to an explicit override within the same constituent, which is why *flightless bird* is unremarkable: *flightless* directly cancels the default

that *bird* would otherwise contribute. Weak entailments behave differently again. When two weak entailments align in direction, as with *utterly devastated*, where both *utterly* and the negative valence of *devastated* point the same way, the combination reads as natural reinforcement. When they conflict, as with *cause celebration* or *foster resentment*, the combination remains fully acceptable, but the tension between the conflicting weak entailments produces a mild rhetorical markedness — the hearer notices that this is an atypical instance of causing or fostering, without experiencing anything approaching contradiction. This pattern of graded markedness, present for weak entailments and absent for strong ones, is itself evidence that the three entailment types are not merely three points arbitrarily marked off on a continuum but correspond to a real difference in how the corresponding features are integrated into processing — a difference explored further in the following two sections.

Table 1 summarizes the correspondence between conventionalization level and entailment type developed in this section, together with the diagnostic that distinguishes each type: whether denial produces contradiction, is possible but marked, or is possible without remark.

Table 1

Conventionalization Levels and Entailment Types

Level	Approximate α	Entailment type	Cancellation behavior	Example
Reference	near 1.0	Strong	Contradictory	<i>dog</i> $\rightarrow$ <i>animal</i>
Denotation	above ~ 0.90	Strong	Contradictory	<i>bachelor</i> $\rightarrow$ <i>unmarried</i>
Sense	~ 0.70 – 0.90	Default	Marked but acceptable	<i>bird</i> $\rightarrow$ <i>flies</i>
Cultural connotation	~ 0.50 – 0.70	Weak	Freely cancellable	<i>regime</i> $\rightarrow$ <i>oppressive</i>
Private connotation	below ~ 0.50	None	Not applicable	<i>dog</i> $\rightarrow$ <i>childhood memory</i>

Cultural Versus Private Connotation

The entailment typology developed above depends on a distinction that prior treatments of connotation have persistently failed to draw: between associations that are systematic at the level of the speech community and associations that are not. Lyons (1977) characterizes connotative meaning broadly as the communicative value an expression carries beyond its conceptual content, and this characterization, taken at face value, folds together phenomena that differ fundamentally in their conventionalization. The systematic tendency of *regime* to collocate with authoritarian governance is not the same kind of thing as one speaker's private association of *hospital* with a parent's illness, even though both are naturally described as "connotative." The

first shows real cross-speaker consistency and real distributional signature; the second shows neither. Treating them as a single phenomenon has led semantic theory to a conclusion that holds only for the second: that connotation as such falls outside the proper domain of semantics and belongs instead to pragmatics or individual psychology (Cruse, 1986). This conclusion is licensed by private connotation. It is not licensed by cultural connotation, and the failure to separate the two is the more consequential mistake.

Cultural connotation comprises associations at the moderate-agreement region of the gradient, systematic enough to generate real corpus patterns and real priming effects, and learnable from linguistic exposure alone without requiring the personal experience that private connotation depends on. A learner could acquire the association between *regime* and authoritarian governance purely from reading, without ever having lived under an authoritarian government, because the association is carried in the distributional record of usage itself; this is precisely what a private, autobiographical association cannot be, since it is not carried in usage at all. Cultural connotation is also gradient in its own right rather than fixed — the strength of the association weakens in contexts that actively support a different reading, as in "democratic regime based on popular sovereignty," and strengthens in contexts that reinforce it, as in "oppressive regime" — and this contextual modulation is itself evidence that the association is integrated with semantic processing rather than sitting outside it as mere statistical residue.

Private connotation comprises associations below this range, arising from individual experience rather than from patterns of usage, generating no systematic corpus signature and no cross-speaker consistency, and for that reason unrecoverable by any method — corpus-distributional, experimental, or computational — that depends on aggregating evidence across a population. The line between the two is not sharp; some associations are shared within a professional or regional subgroup without being shared across the broader speech community, and these occupy

an intermediate position, systematic within the subgroup and effectively private outside it. What marks the boundary, wherever it falls in a given case, is systematicity: whether the association generates a predictable pattern across speakers or only within a single mind.

Recognizing cultural connotation as a genuine, theorizable middle layer between denotation and private association opens questions that the collapsed treatment of "connotation" as a single category could not even pose. How does cultural connotation relate to presupposition and conventional implicature, two other phenomena that contribute to meaning without affecting truth conditions — do all three occupy different, specifiable regions of the same conventionalization gradient, with presupposition nearer the sense level and cultural connotation lower still? How do weak entailments compose when the words carrying them combine, beyond the reinforcement-versus-markedness pattern sketched in the previous section — do more complex compositional structures produce systematic rather than merely intuitive predictions? Do children acquire cultural connotations through the same statistical-learning mechanism that supports denotation acquisition, only more slowly because the signal is noisier at moderate agreement levels, or does a qualitatively different mechanism intervene? And how do cultural connotations shift over historical time — the amelioration of *nice* from "foolish" to "pleasant," the pejoration of *regime* itself — in ways that a framework indexed to synchronic agreement scores should in principle be able to track as movement along the gradient rather than as an unexplained discontinuity? None of these questions can be posed coherently within a framework that has already decided, by conflating cultural and private connotation, that the entire domain is outside semantics. Section six takes up the first and most consequential application of the distinction: the reanalysis of semantic prosody itself.

Semantic Prosody Reconsidered

Semantic prosody names the tendency of a word to appear consistently in contexts of a particular evaluative character, even when that character forms no part of the word's dictionary definition (Louw, 1993; Sinclair, 1991; Stubbs, 2001). *Cause* skews toward negative complements; *commit* skews toward transgression; *utterly* skews toward negative intensification; *set in* skews toward undesirable onset. The pattern has resisted satisfactory treatment on both sides of the lexical-encyclopedic divide it straddles. Corpus linguists have tended to describe it as an evaluative charge absorbed into a word through repeated co-occurrence with evaluatively loaded contexts (Hoey, 2005; Sinclair, 1991), an account that documents the correlation without explaining why particular words absorb particular evaluations rather than others, or why the resulting associations show the structured, feature-based character they do rather than arbitrary co-occurrence. Formal semanticists have tended to dismiss the pattern as a fact about discourse preference rather than about meaning (Whitsitt, 2005), an account that struggles to explain why the pattern is so stable across corpora and registers if it reflects nothing more than what speakers happen to choose to talk about.

The gradient framework relocates the phenomenon rather than choosing between these two accounts. Semantic prosody is cultural connotation, operating at the moderate-agreement region of the gradient and generating weak entailments, and the apparently paradoxical properties that have made it resistant to treatment — systematic yet largely inaccessible to conscious reflection, statistically robust yet resistant to the classical tests of contradiction and anomaly — are exactly what should be expected of a feature at that region of the gradient rather than symptoms of some deeper theoretical confusion. A feature with moderate α is conventionalized enough to shape processing automatically, which produces the systematicity; it is not conventionalized enough to reach the threshold required for reliable conscious articulation,

which produces the intuition-resistance; and it generates probabilistic rather than categorical constraints on usage, which is why classical entailment diagnostics — asking whether denial produces contradiction — simply fail to register it, since weak entailments were never meant to pass that test in the first place.

Locating the phenomenon on the gradient, however, leaves open exactly the question that the absorption account and the dismissal account both fail to answer well: what determines which words show which prosodies, and by what procedure can a given collocational skew be diagnosed as reflecting a genuine weak entailment rather than a discourse-level accident? A companion paper develops this diagnostic in full (Mozahid, 2026, manuscript in preparation), and its core apparatus is worth setting out here directly, since it is what makes the reanalysis proposed in this section falsifiable rather than merely redescriptive.

The apparatus rests on distinguishing two independent readouts of a word's semantic profile, rather than treating the profile as a single evaluatively loaded bundle. A word's *structural axis* comprises its non-evaluative descriptive content — for *cause*, the features of direct, forceful agentivity that make the word apt for describing one thing bringing about another through active intervention rather than gradual development. A word's *valence axis* comprises the conventionalization of evaluative content specifically — whether speakers associate the word itself with positive or negative outcomes, independent of its descriptive content. These two axes can be conventionalized to very different degrees for the same word, and semantic-prosody research has, on this diagnosis, systematically misread skew that belongs to the structural axis as though it were evidence about the valence axis. *Cause* is not a word whose valence has been contaminated by repeated exposure to negative complements; it is a word whose structural profile — direct, forceful, agentive bringing-about — is simply better suited to describing the imposition of unwelcome states than the gradual cultivation of welcome ones, because forceful

imposition is what we more often have occasion to describe when the outcome is harm. The skew is real and the corpus evidence documenting it is not in question; what is in question is which axis it should be attributed to.

This distinction earns its keep only if it yields a procedure for telling, in a given case, which axis a documented skew belongs to, and the procedure proposed for this purpose is the *contradiction test*. For a candidate feature f proposed as part of a word's high-agreement structural profile, the test asks whether the word's attested distribution contains a structured, non-marginal class of uses in which f fails to hold without producing anomaly. If such a class exists and is not itself marked or exceptional, the feature does not belong to the high-agreement structural core; if no such class exists, the feature does belong there. Applied to *cause*, *utterly*, *commit*, *set in*, *foster*, and *regime*, the test consistently locates evaluative valence at the low-to-moderate end of the gradient, beneath a high-agreement structural core that constrains usage independently of valence — *cause celebration*, *utterly delighted*, *foster resentment* are all structured, unmarked counterexamples to any claim that these words carry a high-agreement negative or positive valence, even though each remains fully consistent with the word's structural profile. *Murder*, examined specifically as a contrast case, behaves differently under the same test: no comparably structured, unmarked class of negatively-valenced-yet-non-culpable uses exists, and its valence should accordingly be diagnosed as fused with its descriptive content at or near ceiling agreement, rather than separable from it in the way *cause*'s valence is. The companion paper develops an independent behavioral check on the same diagnosis — a euphemism test, on the reasoning that speakers reach for euphemistic substitution only where valence is genuinely conventionalized rather than merely typical — and a further prediction that the magnitude of a word's collocational skew should not correlate with the conventionalization level of its valence feature, since skew on this account is generated primarily by the interaction of a high-agreement

structural profile with discourse-level asymmetries in what speakers have occasion to describe, not by the strength of an evaluative association per se; checked informally against the six classical cases, skew magnitude and valence conventionalization do in fact decouple in the way this predicts, a decorrelation that a straightforward absorption account, on which stronger skew should simply mean a more deeply absorbed evaluative charge, gives no reason to expect.

This reanalysis resolves the puzzles that have made semantic prosody resistant to treatment, and does so by relocating rather than dissolving them. That the pattern is systematic yet largely unavailable to conscious reflection follows from valence sitting at moderate rather than high conventionalization: sufficient to shape processing automatically, insufficient to support the kind of explicit metalinguistic access that high-agreement features typically permit. That the pattern is statistically robust yet not arbitrary follows from its source in a genuinely structured, high-agreement feature — the descriptive, non-evaluative core of the word's profile — interacting with independently motivated discourse asymmetries, rather than from an unexplained accident of corpus composition. That the pattern resists the classical diagnostics of contradiction and anomaly follows directly from its status as a weak rather than strong entailment, exactly as section four predicts: weak entailments were never expected to behave like strong ones under a contradiction test aimed at valence, which is precisely why a different test, aimed at locating which axis a feature belongs to, was needed in the first place. And that the pattern varies across languages and historical periods in ways that pure semantic universals cannot easily accommodate follows from treating both the structural profile and the valence weighting as independently conventionalized quantities, each free to shift or to diverge cross-linguistically at its own rate.

The relevance of large language models to this diagnosis deserves a brief and appropriately cautious note, since the same evidence has already been put to overreaching use

elsewhere in the literature on distributional learning. Models trained predominantly on political text nonetheless generalize *regime* correctly to *exercise regime* and *dietary regime*, registers in which the valence association common in political usage is largely absent, and models readily produce both *cause damage* and *cause excitement* despite skewed training frequencies for each. This is compatible with the structural-axis account, since a genuinely structural, register-general profile is exactly what should transfer this way; it is not, on its own, sufficient to establish that a model has represented anything resembling a discrete structural-versus-valence distinction rather than having simply encountered enough distributional variance during training to reproduce it without abstracting the underlying structure at all. Distinguishing those two explanations is a question about model internals that behavioral output alone cannot settle (Søgaard, 2022; Potts, 2019), and nothing in the present argument depends on settling it. What the evidence does establish, more modestly, is that the distributional record is rich enough to support profile-consistent generalization by whatever mechanism a given system uses to exploit it — which is itself evidence against treating collocational skew as containing only information about evaluative polarity and nothing more, since a system exploiting only polarity information would have no basis for the structurally-governed pattern of flexibility these models in fact display.

Epistemological Applications: The Gradient and Argument Dynamics

The same gradient that organizes lexical meaning organizes the dynamics of argument, and the connection is not an analogy imposed from outside but a direct consequence of the entailment typology already developed. Whether an argument conducted in a given term converges toward consensus, stalls in impasse, or actively widens the distance between the parties depends on the conventionalization level of the term's contested features, because that level determines what kind of entailment the parties are relying on when they reason from

premises to conclusions, and a shared strong entailment supports a kind of persuasion that a merely local, community-specific agreement does not.

At the top of the gradient, where a feature's conventionalization approaches denotation or reference, argument functions the way classical epistemology assumes it should. Two parties disputing whether a given creature is a dog share the meaning of *dog*; what remains in dispute is a fact about the creature, not a fact about the word, and evidence resolves it. The same holds, at one remove, for disputes conducted in technical vocabulary within mature scientific communities — whether a given particle counts as an electron, whether a given molecule counts as a protein — because terms like these are conventionalized to a very high degree within the community that uses them, and a strong entailment shared by both parties licenses one party persuading the other simply by establishing the premise. This is not a claim about scientific method being superior to other forms of inquiry; it is a claim about what a high-agreement vocabulary makes available that a lower-agreement vocabulary does not, namely a shared inference rule that neither party needs to negotiate before the substantive dispute can even begin.

At the sense level, a different and more limited failure appears. Consider the familiar dispute over whether a tomato is a fruit or a vegetable: both parties understand both the botanical criterion, organized around reproductive structure, and the culinary criterion, organized around taste and use, and each is coherent on its own terms. Neither party is wrong about what the word means; each is prioritizing a different, independently legitimate organization of the same semantic field, and no amount of further evidence about tomatoes moves the dispute, because the dispute was never about tomatoes. The same structure recurs in more consequential form in disputes over whether an act of civil disobedience is *legal* — a dispute between organizing the concept around formal institutional criteria and organizing it around considerations of moral legitimacy, with both organizations independently coherent within the vocabulary of

jurisprudence. What marks disputes at this level is that they reach impasse without polarizing: neither party typically concludes that the other is arguing in bad faith, because each recognizes, at least implicitly, that the disagreement concerns how to organize the field rather than what is true within it. This is the register in which philosophy characteristically operates, and it is worth noting explicitly, since the observation explains a pattern that is otherwise puzzling: philosophical disputes over *knowledge*, *justice*, or *causation* persist for generations not because philosophers reason worse than scientists but because the core vocabulary of philosophy tends to sit at the sense level rather than the denotation level — agreement high enough to sustain genuine communication, low enough that competing organizations of the same conceptual field can each claim real support. Scientific vocabulary converges because mature sciences maintain, through training and institutional correction, technical terms conventionalized well above the sense level; philosophical vocabulary does not converge in the same way not because philosophers fail at consensus-formation but because the object of philosophical inquiry is very often the organization of a concept itself, which is exactly the kind of dispute a sense-level vocabulary cannot settle from within.

Below the sense level, at the region of cultural connotation, a further and more consequential dynamic appears, and it is here that argument does not merely stall but actively widens the gap between the parties. Disputes over *freedom* in political discourse are the clearest illustration: the term carries very high agreement at the level of bare denotation — freedom is, roughly, absence of constraint — but sharply diverging conventionalization at the level of which constraints are salient, with different political traditions treating freedom from government interference, freedom from economic necessity, or freedom from social domination as the operative sense. What distinguishes this case from the tomato case, and what explains why it polarizes rather than merely stalling, is not simply that the agreement score is lower. It is that

each party in the freedom dispute is embedded within a subcommunity — a political tradition, a professional or ideological formation — within which the contested feature is itself highly conventionalized. A committed libertarian and a committed socialist each experience their own conception of freedom as obvious, shared, and virtually undeniable, because within their respective communities it is exactly that. What neither party has occasion to notice, from inside the dispute, is that the high agreement they are each drawing on is local rather than general: a feature can carry near-denotational agreement within a subcommunity while carrying only cultural-connotation-level agreement across the population as a whole, and the experience of holding a locally high-agreement belief is indistinguishable, from the inside, from holding a genuinely high-agreement one.

This is the mechanism I call *alpha misattribution*: treating a feature whose agreement score is high only within one's own subcommunity as though it carried the stability of a term whose agreement score is high across the general population. The error is not a failure of reasoning about the substantive question in dispute; it is a failure to correctly diagnose the semantic status of the term the dispute is being conducted in. Its consequence is specific and predictable. A party relying on a strong entailment, when that entailment is not in fact shared by the interlocutor, does not conclude that the interlocutor organizes the concept differently; a strong entailment is not experienced as one organization among others, but as simply true, so the interlocutor's failure to accept it reads not as disagreement but as resistance to an obvious inference — as evasion, bias, or bad faith. This is what separates polarization from mere impasse. The tomato dispute does not polarize because neither party mistakes their organization of *fruit* for a fact the other side is willfully denying. The freedom dispute polarizes because each party experiences its own conception as exactly that kind of fact, and interprets its absence in the interlocutor as a moral or intellectual failing rather than as evidence of an unresolved population-

level disagreement. The dynamic, once triggered, is largely self-reinforcing: each attribution of bad faith reduces the interlocutor's actual openness to persuasion, which is then read as further confirmation of the original attribution, entrenching a gap that no exchange of evidence about the substantive question was ever going to close, because the substantive question was never where the disagreement lived.

The abortion debate is the paradigm case of this pattern, and it rewards examining directly because it isolates the mechanism with unusual clarity. The central contested term is standardly identified as *person* or *human being*, but this identification obscures more than it reveals, because it treats the whole term as contested when in fact only part of it is. At the level of basic biological and genetic membership — that the entity in question is a living human organism, genetically continuous with the adults who produced it — agreement is very high and effectively uncontested; whatever divides the two sides of this debate, it is not disagreement about the biology at that level. The actual site of disagreement is a distinct feature, the *moral-status-onset* question of when, if at any determinate point, an entity of that biological kind acquires the properties that ground moral status and the protections that follow from it. This is worth separating explicitly from a further question sometimes run together with it, the biological question of precisely when organismal life begins in a developmental sense; that question involves its own genuine indeterminacies — the phenomenon of twinning, the gradual and non-instantaneous character of fertilization — but those indeterminacies are properly a matter of Type 1 vagueness in the sense developed earlier in this paper, uncertainty about whether a shared, high-agreement criterion applies to a borderline case, and not evidence of low agreement on the criterion itself. The genuinely low-agreement term in this debate is moral-status-onset, not organismal-life-onset, and treating the two as a single contested question, as the ordinary framing of the debate does, obscures exactly the distinction the gradient framework is built to draw.

On moral-status-onset, each side of the debate occupies a position that is highly conventionalized within its own community and only weakly conventionalized across the population as a whole, which is exactly the structural precondition for misattribution developed above. One tradition organizes moral status around genetic and biological continuity from conception onward, and within that tradition the criterion functions with something close to denotational certainty. Another organizes moral status around the gradual acquisition of psychological capacities — sentience, the beginnings of consciousness — and within that tradition its own criterion functions with comparable certainty. Neither community is wrong about what it, internally, agrees on; what each community lacks, from the inside, is any signal that its internal agreement does not extend to the population that the debate is actually conducted in front of. The result is exactly the pattern predicted: arguments that talk past each other, since each side is arguing from a premise the other does not share and does not experience as merely one premise among available alternatives; accumulating biological and neuroscientific evidence that fails to move the dispute, since that evidence is absorbed into each side's existing framework rather than adjudicating between them; and entrenchment over time, since each side's failure to persuade the other is read, from within the misattribution, as confirmation that the other side is arguing in bad faith rather than as evidence that the underlying term is not yet, and may never be, conventionalized across the wider population. The normative implication that follows is deliberately narrow and epistemic rather than moral. What the framework recommends is a form of calibration: that confidence in the prospect of resolving a dispute through further argument should track the actual, population-level conventionalization of the term the dispute is conducted in, and that accepting a term as currently low-agreement is a claim about the present state of a speech community's semantic convergence, not a claim about whether the underlying moral question has a stance-independent answer. A committed realist about moral status and a

committed anti-realist can both accept, without inconsistency, that the term *person* in its morally loaded sense is not currently conventionalized across the relevant population; the recommendation to calibrate confidence accordingly is compatible with either metaethical position and asserts neither.

That this pattern is not an artifact of political-debate specifically, invented to explain one especially fraught dispute, is worth establishing independently, and legal interpretation supplies exactly this kind of independent case. Judges routinely disagree about whether a given piece of statutory or contractual language is clear, and the disagreement is not confined to hard cases; Justice Kavanaugh has argued at length that judges lack any shared, principled standard for determining clarity in the first place, and that determinations of clarity in practice reduce to something closer to individual judicial instinct dressed in doctrinal language (Kavanaugh, 2016, as discussed in Choi, 2024). Hart's own illustration — whether a bicycle counts as a *vehicle* under a sign prohibiting vehicles from a park — has been taken up in recent empirical work measuring what Choi (2024) calls the zone of indeterminacy: the region in which a term's application to a given case is genuinely unsettled rather than merely under-evidenced. What that empirical work finds is instructive for present purposes regardless of its own conclusions about legal method: interpreters with strong textualist commitments tend to report narrower zones of indeterminacy than the underlying data supports, systematically treating language as determinately clear in cases where, by the interpreters' own stated standards, no principled basis for that determination exists. This is alpha misattribution in a domain with no obvious connection to the political case: a professional community's internal confidence in a criterion — that the plain meaning of a text settles a given case — outrunning what that criterion can actually deliver once tested against genuinely contested applications, with disagreement among judges read, as a

result, as one side failing to follow the plain text rather than as evidence that the text's meaning was not, on the point in question, conventionalized to begin with.

The pattern documented in each of these domains is best explained by two levels operating together rather than by either alone. The gradient framework, developed at the level of semantics and cognition, explains the pathology itself — why arguments in low-agreement vocabulary talk past each other and why misattribution converts that impasse into polarization. It does not, on its own, explain why particular disputes are sustained and amplified at the institutional level over long periods, and a full account of that sustaining mechanism is beyond this paper's scope; it is enough to note here that institutions and mobilization efforts built around a dispute have a structural incentive to preserve rather than correct the underlying misattribution, since a narrative in which the opposing side is withholding a resolution in bad faith organizes political and legal action more effectively than a narrative in which the dispute rests on an unresolved and perhaps currently unresolvable semantic disagreement. The two levels do not reduce to one another, and developing the interaction between them is a task for future work rather than for the present paper.

Two further implications follow for how metalinguistic negotiation — explicit discussion about what a contested term should mean — succeeds or fails. Negotiation tends to succeed when the parties can appeal to shared higher-order principles for adjudicating between their organizations of a concept, or when institutional mechanisms exist that can raise a term's conventionalization through repeated, enforced use; legal terms such as *contract* or *tort* illustrate the latter path directly, having achieved high conventionalization within legal communities specifically through sustained institutional reinforcement (Choi, 2024) in a way that terms outside any comparable institutional structure rarely do. Negotiation tends to fail when the contested feature has genuinely low population-level agreement and the parties' commitments to

their respective organizations are bound up with group identity in a way that makes conceding costly independent of the semantic merits — the condition that obtains, on the diagnosis given above, in both the abortion debate and in the more polarized reaches of legal and political interpretation generally. Recognizing which condition obtains in a given dispute is itself, on this account, the first and most useful thing rational participants in that dispute can do.

Predictions and Future Directions

The framework developed here is intended as a theoretical structure with precise, testable consequences, not as a redescription immune to disconfirmation, and this section states the consequences explicitly rather than leaving them implicit in the preceding argument.

If conventionalization is genuinely continuous rather than clustering at the five reference points used for exposition, then large-scale agreement studies measuring feature-word associations across diverse semantic domains should find agreement scores distributed continuously rather than bimodally, with no natural gap separating a lexical cluster from an encyclopedic one. If entailment strength tracks conventionalization level as section four proposes, then acceptability-judgment studies using cancellation frames — of the form "*w* but not *f*" — should find acceptability ratings correlating with independently measured agreement scores in a consistently monotonic direction, without commitment, on the present evidence, to any particular functional shape for that relationship. If default and weak entailments are genuinely active during processing even when ultimately cancelled, priming studies should continue to find facilitation effects for cancelled defaults, of the kind already documented for *bird* and flight (McRae et al., 1997), extending systematically to weak entailments at the cultural-connotation level, where the same dissociation between behavioral effect and cancellability is predicted to hold even more starkly, since the underlying agreement score is lower still.

The structural-axis and valence-axis distinction developed in section six generates its own predictions independently of the broader gradient apparatus. The decorrelation between skew magnitude and valence conventionalization, checked informally against six cases in the companion paper (Mozahid, 2026, manuscript in preparation), should hold under systematic norming across a much larger sample of prosodically marked words; a scalar absorption account predicts a positive correlation between these two quantities, and the gradient account predicts none, which makes this a genuine point of empirical adjudication between the two rather than a matter of interpretive preference.

The argument-dynamics application generates a prediction with a specific and testable structure, extending directly from the diagnosis of alpha misattribution in section seven. If polarization at the cultural-connotation level is driven specifically by participants mistaking a subcommunity-level agreement score for a population-level one, then an intervention that makes this gap explicit — informing participants, before or during a structured discussion, of the documented divergence between within-community and population-wide agreement on the contested term — should reduce post-discussion belief divergence relative to an uninformed control condition, but should do so specifically for disputes conducted in cultural-connotation-level vocabulary. The same intervention applied to a sense-level dispute, of the tomato-and-fruit variety, should show little or no comparable effect, because impasse at that level is not driven by misattribution in the first place; the parties to a sense-level dispute are not, on this account, mistaking a local agreement score for a general one, so correcting that mistake has nothing to act on. This differential prediction — an effect at the cultural-connotation level and its absence at the sense level — is what would make such a study a genuine test of the misattribution mechanism specifically, rather than a demonstration of the weaker and less interesting claim that any metacognitive framing reduces interpersonal conflict. A finding of comparable reduction at both

levels would count as evidence against misattribution as the specific mechanism driving polarization, and in favor of some more general debiasing effect operating independently of it; the prediction is stated with that failure condition attached, rather than framed so as to be confirmed under either outcome.

Limitations and Future Work

Several limitations should be acknowledged directly rather than left for a reviewer to discover. The measurement of α remains, at present, a theoretical commitment rather than an accomplished program: the values used illustratively throughout this paper are estimates offered for their plausibility, not findings, and the paper's own claims about where particular features fall on the gradient stand or fall on norming and corpus work that remains to be done. The compositional behavior of weak entailments has been characterized here only at the level of reinforcement and markedness for simple cases; a fuller compositional semantics, extending to more complex constructions, is a substantial undertaking left for future work. The relationship between the population-level and subcommunity-level agreement scores that does the central work in the misattribution diagnosis has been characterized qualitatively; a more precise account of how subcommunity boundaries themselves should be drawn, and of how agreement scores computed relative to different candidate boundaries relate to one another, would strengthen the diagnostic considerably. And the two-level explanation offered for why misattribution is sustained rather than corrected at the institutional level — that political and legal mobilization efforts have structural incentive to preserve rather than resolve the underlying confusion — has been stated here only in outline; developing the mechanism by which that incentive actually operates, and testing it against the historical record of specific disputes, is a project distinct from the present one.

Conclusion

The relationship between lexical meaning and world knowledge has resisted a stable categorical boundary because no such boundary exists to be found; what theory has been approximating, across a century of disagreement about where the line falls, is a gradient of conventionalization, indexed by the degree of agreement a semantic feature commands across a speech community. Locating meaning on this gradient rather than on either side of a boundary dissolves the instability documented in section two, supplies the entailment typology needed to represent inference at every point on the scale rather than only at its top, and diagnoses the conflation of cultural and private connotation that has kept a genuinely systematic phenomenon — cultural connotation, and the weak entailments it generates — outside the reach of semantic theory for far longer than the evidence has warranted. Applied to semantic prosody, the framework relocates rather than dissolves a set of puzzles that have resisted treatment for three decades, and does so with the aid of a genuine diagnostic, the contradiction test, for telling which axis of a word's profile a given collocational skew belongs to.

The same gradient, applied to argument rather than to lexical items, explains a pattern that would otherwise require separate and unconnected accounts: why disputes conducted in high-agreement vocabulary converge, why disputes conducted in moderate-agreement vocabulary sometimes merely stall and sometimes actively widen the distance between the parties, and why the difference between those two outcomes turns not on the bare agreement score alone but on whether the parties have mistaken a locally high agreement for a generally high one. Alpha misattribution, diagnosed this way, is neither a moral failing nor a failure of reasoning about the substantive matter in dispute; it is a failure to correctly diagnose the semantic status of the term the dispute is conducted in, and it recurs, on the evidence considered here, in domains as different as the abortion debate and statutory interpretation. What the

framework recommends in response is calibration rather than resolution: confidence that argument can produce consensus should track the term's actual, current conventionalization, a recommendation that is available to realists and anti-realists alike about whatever underlying question the contested term concerns, because it is a claim about the present state of a speech community's semantic convergence and nothing more.

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